EXPERIMENT 2 (HARIN T_192412412)

1. Aim

To analyze the reflection and transmission characteristics of a microwave transmission line using MATLAB by calculating return loss, insertion loss, and transmission efficiency.

2. Apparatus Required

1. **Computer or laptop**
2. **MATLAB software**

3. Procedure

1. **Open MATLAB and create a new script.**
2. **Define the characteristic impedance, load impedance, line length, and frequency range.**
3. **Calculate the reflection coefficient, \(S_{11}\), and transmission coefficient, \(S_{21}\).**
4. **Determine the return loss and insertion loss over the frequency range.**
5. **Plot the reflection and transmission magnitudes in dB.**
6. **Plot the return loss and insertion loss against frequency.**
7. **Observe the graphs and analyze the transmission-line performance.**

4. MATLAB Program

```
clc; clear; close all;
c=3e8; Z0=50; ZL=50; L=0.05;
F=linspace(8e9,12e9,201);
beta=2*pi*F/c;
G=(ZL-Z0)/(ZL+Z0);
S11=G*ones(size(F)); S22=S11;
S21=exp(-1j*beta*L); S12=S21;
RL=-20*log10(max(abs(S11),1e-10));
IL=-20*log10(abs(S21));
eta=abs(S21).^2*100;
fprintf('HARIN | 192412412\n');
fprintf('Wavelength = %.1f mm\n',c/10e9*1e3);
```

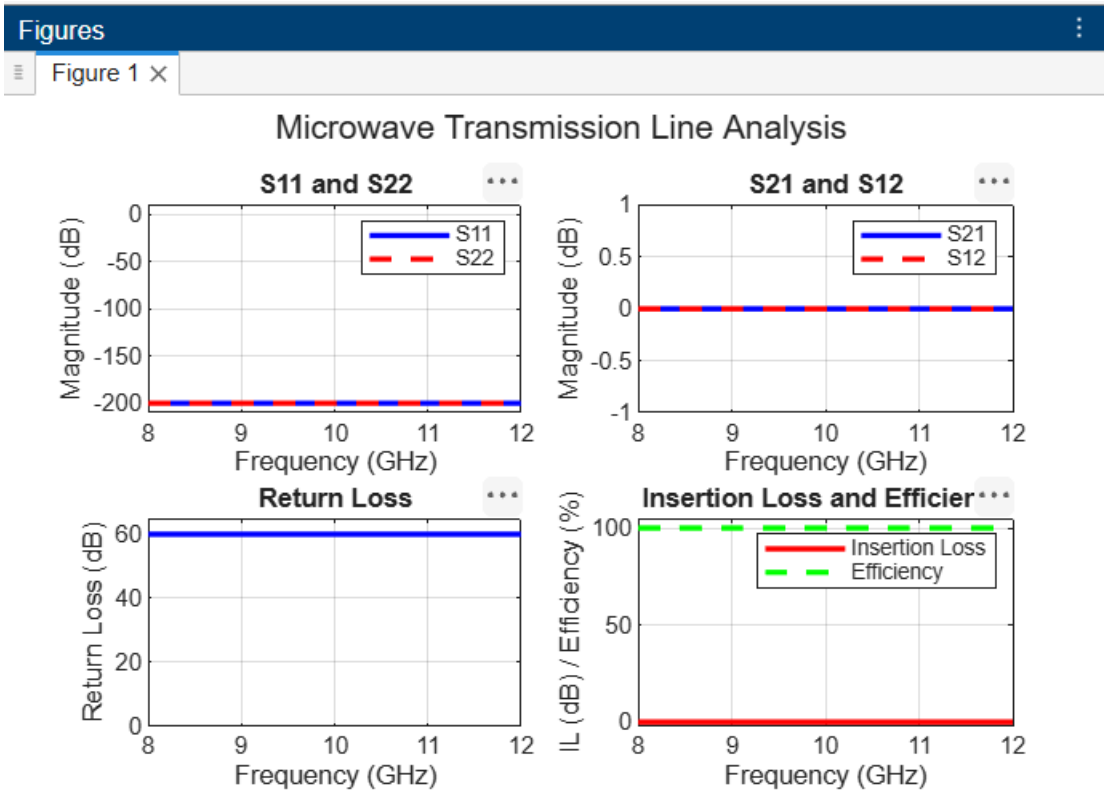
```
fprintf('Beta = %.2f rad/m\n',2*pi*10e9/c);
fprintf('Return Loss = Inf dB\n');
fprintf('Insertion Loss = %.2f dB\n',IL(101));
fprintf('Efficiency = %.1f%%\n',eta(101));
figure('Color','w','Position',[100 100 1000 650]);
subplot(2,2,1);
plot(F/1e9,20*log10(max(abs(S11),1e-10)),'b','LineWidth',2);
hold on;
plot(F/1e9,20*log10(max(abs(S22),1e-10)),'r--','LineWidth',2);
grid on; xlim([8 12]); ylim([-210 10]);
xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)');
title('S11 and S22'); legend('S11','S22');
subplot(2,2,2);
plot(F/1e9,20*log10(abs(S21)),'b','LineWidth',2);
hold on;
plot(F/1e9,20*log10(abs(S12)),'r--','LineWidth',2);
grid on; xlim([8 12]); ylim([-1 1]);
xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)');
title('S21 and S12'); legend('S21','S12');
subplot(2,2,3);
plot(F/1e9,min(RL,60),'b','LineWidth',2);
grid on; xlim([8 12]); ylim([0 65]);
xlabel('Frequency (GHz)'); ylabel('Return Loss (dB)');
title('Return Loss');
subplot(2,2,4);
plot(F/1e9,IL,'r','LineWidth',2);
hold on;
plot(F/1e9,eta,'g--','LineWidth',2);
grid on; xlim([8 12]); ylim([-2 105]);
xlabel('Frequency (GHz)'); ylabel('IL (dB) / Efficiency (%)');
```

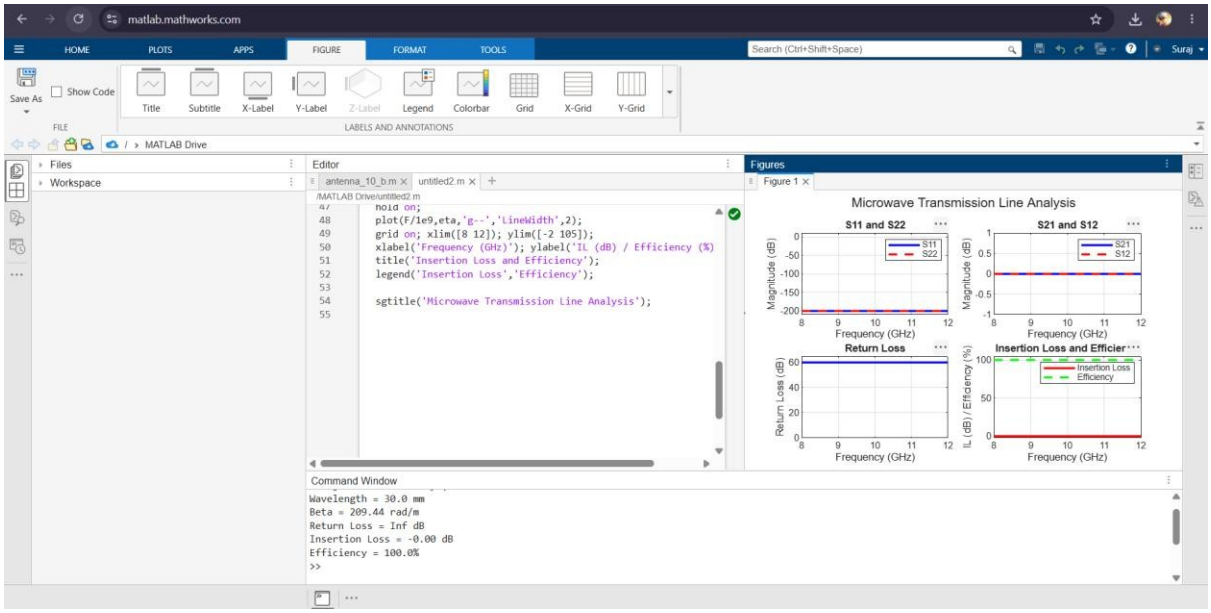
title('Insertion Loss and Efficiency');

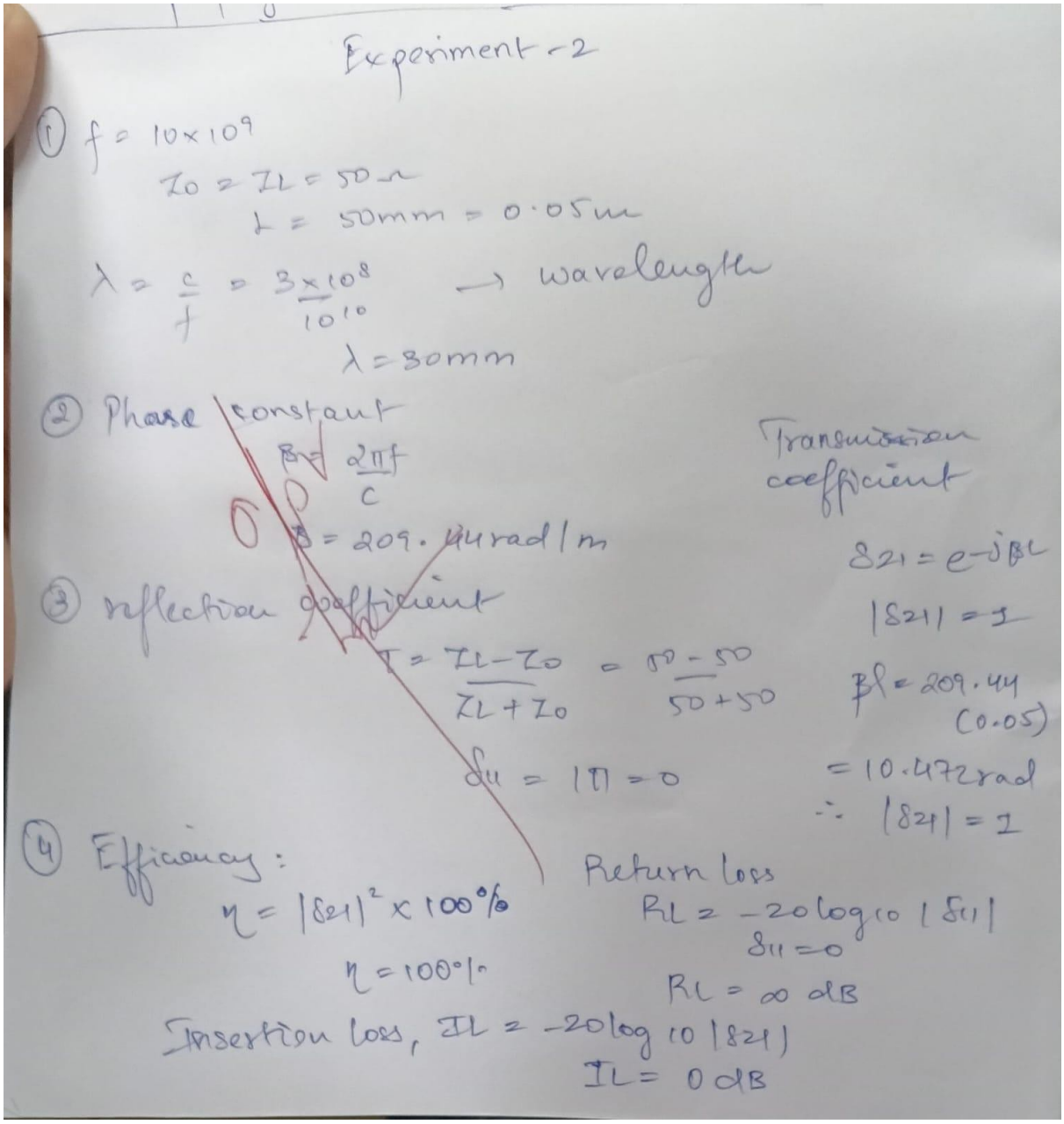
legend('Insertion Loss','Efficiency');

sgtitle('Microwave Transmission Line Analysis');

OUTPUT:







5. Reflection

I learned how to analyze reflection and transmission characteristics in a microwave transmission line using MATLAB. I understood the significance of S-parameters, return loss, insertion loss, and impedance matching in determining transmission-line performance.

6. Result

The reflection and transmission characteristics of a 50 Ω matched microwave transmission line were analyzed using MATLAB. The graphs of S_{11}, S_{21}, return loss, and insertion loss were obtained. For the ideal matched, lossless line, the reflection coefficient is zero, return loss approaches infinity, and insertion loss is zero dB.